

CREATE TABLE STATEMENTS

```
CREATE TABLE GENE (  
  gene_id          INT          NOT NULL      AUTO_INCREMENT,  
  gene_symbol      VARCHAR(20)  NOT NULL,  
  short_description VARCHAR(500),  
  chromosome       VARCHAR(20)  NOT NULL,  
  start_position   INT          NOT NULL,  
  end_position     INT          NOT NULL,  
  strand          CHAR(1)       NOT NULL,  
  organism_id      int         NOT NULL,  

```

```
CONSTRAINT GENE_PK  
  PRIMARY KEY (gene_id)  
);
```

```
CREATE TABLE ORGANISM (  
  organism_id      INT          NOT NULL      AUTO_INCREMENT,  
  scientific_name   VARCHAR(100) NOT NULL,  
  domain           VARCHAR(8)   NOT NULL,  
  genome_size      BIGINT,  
  chromosome_count INT,  
  taxonomy_id      INT,  

```

```
CONSTRAINT ORG_PK  
  PRIMARY KEY (organism_id)  
);
```

```
CREATE TABLE GENE_PRODUCT (  
  gene_product_id  INT          NOT NULL      AUTO_INCREMENT,  
  gene_id          INT          NOT NULL,  
  product_name     VARCHAR(100) NOT NULL,  
  product_type     VARCHAR(7)   NOT NULL,  
  description       VARCHAR(500),  

```

```
CONSTRAINT GENEPRO_PK  
  PRIMARY KEY (gene_product_id)  
);
```

```
CREATE TABLE GENE_DISEASE (  
  disease_id       INT          NOT NULL      AUTO_INCREMENT,  
  disease_name     VARCHAR(150) NOT NULL,  
  category         VARCHAR(15)  NOT NULL,  

```

```
CONSTRAINT GENEDISEASE_PK  
  PRIMARY KEY (disease_id)  
);
```

```
CREATE TABLE GENE_DISEASE_ASSOCIATION (  
  gene_id          INT          NOT NULL,  
  disease_id       INT          NOT NULL,  

```

```
CONSTRAINT GDA_PK  
  PRIMARY KEY (gene_id, disease_id)  
);
```

```
CREATE TABLE GENE_PATHWAY (
    pathway_id          INT          NOT NULL      AUTO_INCREMENT,
    pathway_name        VARCHAR(150) NOT NULL,
    description          VARCHAR(500),
```

```
CONSTRAINT GENEPATHWAY_PK
    PRIMARY KEY (pathway_id)
);
```

```
CREATE TABLE GENE_PATHWAY_ASSOCIATION (
    gene_id             INT          NOT NULL,
    pathway_id          INT          NOT NULL,
```

```
CONSTRAINT GPA_PK
    PRIMARY KEY (gene_id, pathway_id)
);
```

```
CREATE TABLE GENE_FAMILY (
    family_id           INT          NOT NULL      AUTO_INCREMENT,
    family_name          VARCHAR(150) NOT NULL,
    description          VARCHAR(500),
```

```
CONSTRAINT GENEFAMILY_PK
    PRIMARY KEY (family_id)
);
```

```
CREATE TABLE GENE_FAMILY_MEMBER (
    gene_id             INT          NOT NULL,
    family_id           INT          NOT NULL,
```

```
CONSTRAINT GFM_PK
    PRIMARY KEY (gene_id, family_id)
);
```

```
CREATE TABLE GO_TERM (
    go_id               VARCHAR(15)   NOT NULL,
    term_name            VARCHAR(200) NOT NULL,
    ontology              VARCHAR(2)   NOT NULL,
    definition            VARCHAR(500) NOT NULL,
    date_created          DATETIME     NOT NULL,
    date_last_update     DATETIME     NOT NULL,
```

```
CONSTRAINT GOTERM_PK
    PRIMARY KEY (go_id)
);
```

```
CREATE TABLE GO_TERM_RELATION (
    go_relation_id       INT          NOT NULL      AUTO_INCREMENT,
    parent_go_id         VARCHAR(15)   NOT NULL,
    child_go_id          VARCHAR(15)   NOT NULL,
    relation_term         VARCHAR(9)    NOT NULL,
```

```
CONSTRAINT GTR_PK
    PRIMARY KEY (go_relation_id)
);
```

```

CREATE TABLE SOURCE (
source_id          INT          NOT NULL      AUTO_INCREMENT,
source_name        VARCHAR(150) NOT NULL,
publish_year       YEAR         NOT NULL,
url                VARCHAR(2048) NOT NULL,

CONSTRAINT SOURCE_PK
PRIMARY KEY (source_id)
);

CREATE TABLE GENE_ANNOTATION (
annotation_id      INT          NOT NULL      AUTO_INCREMENT,
gene_product_id    INT          NOT NULL,
go_id              VARCHAR(15)  NOT NULL,
evidence_code      VARCHAR(3)   NOT NULL,
date_created       DATETIME     NOT NULL,
date_last_update   DATETIME     NOT NULL,
comment            VARCHAR(1000),

CONSTRAINT GENEANN_PK
PRIMARY KEY (annotation_id)
);

CREATE TABLE ANNOTATION_SOURCE (
source_id          INT          NOT NULL,
annotation_id      INT          NOT NULL,

CONSTRAINT ANNSOURCE
PRIMARY KEY (source_id, annotation_id)
);

```

INSERT INTO STATEMENTS

INSERT INTO ORGANISM (scientific_name, domain, genome_size, chromosome_count, taxonomy_id) #

Surrogate key used, last 3 attributes are not mandatory

VALUES

('Homo sapiens', 'eukarya', 3200000000, 46, 9606),
('Drosophila melanogaster', 'eukarya', 140000000, 8, 7227),
('Arabidopsis thaliana', 'eukarya', 135000000, 10, 3702),
('Escherichia coli', 'bacteria', 4600000, 1, 562),
('Caenorhabditis elegans', 'eukarya', 100000000, 12, 6239),
('Mus musculus', 'eukarya', 2700000000, 20, 10090);

INSERT INTO GENE (gene_symbol, chromosome, start_position, end_position, strand, organism_id)

VALUES

('TP53', '17', 7668421, 7687490, '-', 1),
('EGFR', '7', 55019017, 55211628, '+', 1),
('APOE', '19', 44905796, 44909393, '+', 1),
('TNF', '6', 31575565, 31578336, '+', 1),
('IL6', '7', 22727200, 22731998, '+', 1),
('VEGFA', '6', 43770211, 43786487, '+', 1),
('TGFB1', '19', 41330323, 41353922, '-', 1),
('MTHFR', '1', 11785723, 11805964, '-', 1),
('BRCA1', '17', 43044295, 43170327, '-', 1),
('ACE', '17', 63477061, 63498373, '+', 1), # gene_id = 10
('STAT3', '17', 42313324, 42388442, '-', 1),

('Blm', '3R', 11736810, 11741783, '-', 2),
('N', 'X', 3134870, 3172221, '+', 2),
('dpp', '2L', 2428372, 2459823, '+', 2),

('PHYB', '2', 8139756, 8144461, '+', 3),
('FLC', '5', 3173382, 3179448, '-', 3),
('PHYA', '1', 3095160, 3100819, '-', 3),

('recA', '1', 2822708, 2823769, '-', 4),
('dnaA', '1', 3882326, 3883729, '-', 4),
('rho', '1', 3966417, 3967676, '+', 4), # gene_id = 20

('daf-16', '1', 10750498, 10776703, '+', 5),
('daf-2', '3', 2994514, 3040846, '-', 5),
('skn-1', '4', 5651039, 5660384, '-', 5),

('BRCA2', '13', 32315077, 32400268, '+', 1),
('PALB2', '16', 23603165, 23641310, '-', 1),

('TRP53', '11', 69471174, 69482699, '+', 6),
('TRP63', '16', 25502513, 25710842, '+', 6),
('TRP73', '4', 154140706, 154224332, '-', 6),
('TP63', '3', 189596746, 189897276, '+', 1);

INSERT INTO GENE_PRODUCT (product_name, product_type, description, gene_id)

VALUES

('tumor necrosis factor', 'protein', 'cytokine', 4),
('ACE protein', 'protein', 'angiotensin I converting enzyme', 10),
('Rho protein', 'protein', 'transcription termination factor', 20),
('BRCA1 protein', 'protein', 'nuclear phosphoprotein', 9),

('BRCA2 protein', 'protein', 'nuclear DNA repair protein', 24),
('PALB2 protein', 'protein', 'partner and localizer of BRCA2 protein', 25),
('phytochrome B', 'protein', 'a plant photoreceptor that forms a photobody', 15),
('FLC protein', 'protein', 'a MADS-box transcription factor that represses floral transition and contributes to temperature compensation of the circadian clock', 16),
('phytochrome A', 'protein', 'a photoreceptor responsible for sensing far-red light', 17);

INSERT INTO GENE_FAMILY (family_name)
VALUES
('p53 family'),
('ErbB family'),
('PDGF/VEGF growth factor family');

INSERT INTO GENE_FAMILY_MEMBER (gene_id, family_id)
VALUES
(1, 1),
(2, 2),
(6, 3),
(26, 1),
(27, 1),
(28, 1),
(29, 1);

INSERT INTO GENE_PATHWAY (pathway_name)
VALUES
('cell cycle pathway'),
('apoptosis pathway'),
('MAPK signaling pathway');

INSERT INTO GENE_PATHWAY_ASSOCIATION (gene_id, pathway_id)
VALUES
(1, 1),
(1, 2),
(1, 3),
(9, 1),
(4, 2),
(4, 3),
(7, 2),
(7, 3),
(5, 3),
(6, 3),
(11, 2),
(11, 3);

INSERT INTO GENE_DISEASE (disease_name, category)
VALUES
('li-fraumeni syndrome', 'genetic'),
('breast cancer', 'cancer'),
('lung cancer', 'cancer'),
('alzheimer's disease', 'neurological'),
('cardiovascular disease', 'cardiovascular'),
('rheumatoid arthritis', 'autoimmune'),
('crohn's disease', 'autoimmune'),
('psoriasis', 'autoimmune');

INSERT INTO GENE_DISEASE_ASSOCIATION (gene_id, disease_id)

VALUES

(1,1),
(1,2),
(1,3),
(3,4),
(3,5),
(4,6),
(4,7),
(4,8),
(24, 2),
(25, 2),
(9, 2);

INSERT INTO GO_TERM (go_id, term_name, ontology, definition, date_created, date_last_update)

VALUES

('GO:0006915', 'apoptotic process', 'bp', 'The process of programmed cell death that occurs when a cell receives an internal or external signal, and proceeds through a sequence of biochemical events which trigger an execution event.', NOW(), NOW()),
('GO:0008283', 'cell proliferation', 'bp', 'The process by which cells grow and divide to create more cells.', NOW(), NOW()),
('GO:0007165', 'signal transduction', 'bp', 'The cellular process where a signal is transmitted from a receptor to elicit a cellular response.', NOW(), NOW()),
('GO:0012501', 'programmed cell death', 'bp', 'A highly regulated process of cellular self-destruction.', NOW(), NOW()),
('GO:0009987', 'cellular process', 'bp', 'Any process that is carried out at the cellular level.', NOW(), NOW()),
('GO:0005634', 'nucleus', 'cc', "A membrane bound organelle in eukaryotic cells that contains all of the cell's chromosomes", NOW(), NOW()),
('GO:0005737', 'cytoplasm', 'cc', "The jelly-like substance found inside a cell", NOW(), NOW()),
('GO:0007154', 'cell communication', 'bp', "Any process by which a cell gives and receive interactions between another cell or its surroundings", NOW(), NOW()),
('GO:0007267', 'cell-cell signaling', 'bp', 'Any process that mediates the transfer of information between cells', NOW(), NOW()),
('GO:0035426', 'extracellular matrix-cell signaling', 'bp', 'Any process that mediates the transfer of information between a cell and the extracellular matrix', NOW(), NOW()),
('GO:0010647', 'positive regulation of cell communication', 'bp', 'A process that increases the rate of cell communication', NOW(), NOW());

INSERT INTO GO_TERM_RELATION(parent_go_id, child_go_id, relation_term)

VALUES

('GO:0009987', 'GO:0007165', 'is a'),
('GO:0009987', 'GO:0008283', 'is a'),
('GO:0012501', 'GO:0006915', 'is a'),
('GO:0007154', 'GO:0007165', 'part of'),
('GO:0007154', 'GO:0007267', 'is a'),
('GO:0007154', 'GO:0035426', 'is a'),
('GO:0007154', 'GO:0010647', 'regulates');

INSERT INTO GENE_ANNOTATION (gene_product_id, go_id, evidence_code, date_created, date_last_update, comment)

VALUES

(1, 'GO:0006915', 'IDA', NOW(), NOW(), 'TNF induces apoptosis via TNFR1 signaling'),
(2, 'GO:0007165', 'IDA', NOW(), NOW(), 'ACE catalyzes angiotensin I to II, which activates signaling'),
(3, 'GO:0009987', 'IDA', NOW(), NOW(), 'Rho protein mediates the termination of transcription in E. coli'),
(7, 'GO:0005634', 'IDA', NOW(), NOW(), 'PHYB translocated to the nucleus in response to light'),
(8, 'GO:0005634', 'EXP', NOW(), NOW(), 'FLC functions as a nuclear transcription factor'),
(9, 'GO:0005737', 'IEA', NOW(), NOW(), 'PHYA is localized in the cytoplasm prior to being activated');

```

INSERT INTO SOURCE (source_name, publish_year, url)
VALUES
('Reactive Oxygen Species in TNF $\alpha$ -Induced Signaling and Cell Death', 2010,
'https://www.sciencedirect.com/science/article/pii/S1016847823109988'),
('Physiology, Renin Angiotensin System', 2026, 'https://www.ncbi.nlm.nih.gov/books/NBK470410/'),
('Regulation of Rho-Dependent Transcription Termination by NusG Is Specific to the Escherichia coli Elongation
Complex', 2000, 'https://pubs.acs.org/doi/10.1021/bi992658z'),
('Interaction with plant transcription factors can mediate nuclear import of phytochrome B', 2012,
'https://www.pnas.org/doi/10.1073/pnas.1120764109'),
('Light-Regulated Nuclear Import and Degradation of Arabidopsis Phytochrome-A N-Terminal Fragments', 2010,
'https://pmc.ncbi.nlm.nih.gov/articles/PMC3037077/'),
('UniProt', 2024, 'https://www.uniprot.org/uniprotkb/Q9S7Z8/entry');

```

```

INSERT INTO ANNOTATION_SOURCE (source_id, annotation_id)
VALUES
(1, 1),
(2, 2),
(3, 3),
(4, 4),
(5, 5),
(6, 6);

```

ALTER TABLE STATEMENTS

ALTER TABLE GENE

ADD CONSTRAINT GENE_ORG_FK

FOREIGN KEY (organism_id) REFERENCES ORGANISM (organism_id)

ON DELETE RESTRICT ON UPDATE CASCADE,

ADD CONSTRAINT GENE_SYMBOL_UNIQUE

UNIQUE (gene_symbol, organism_id),

ADD CONSTRAINT GENE_START_CHECK

CHECK (start_position >= 0 AND start_position < end_position),

ADD CONSTRAINT GENE_STRAND_CHECK

CHECK (strand IN ('+', '-'));

ALTER TABLE ORGANISM

ADD CONSTRAINT ORG_SCI_UNIQUE

UNIQUE (scientific_name),

ADD CONSTRAINT ORG_DOMAIN_CHECK

CHECK (domain IN ('bacteria', 'archaea', 'eukarya'));

ALTER TABLE GENE_PRODUCT

ADD CONSTRAINT GENEPRO_GENE_FK

FOREIGN KEY (gene_id) REFERENCES GENE (gene_id)

ON DELETE RESTRICT ON UPDATE CASCADE,

ADD CONSTRAINT GENEPRO_PROTYPE_CHECK

CHECK (product_type IN ('dna', 'rna', 'protein', 'other'));

ALTER TABLE GENE_DISEASE

ADD CONSTRAINT GENEDISEASE_NAME_UNIQUE

UNIQUE (disease_name),

ADD CONSTRAINT GENEDISEASE_CATEGORY_CHECK

CHECK (category IN ('genetic', 'infectious', 'metabolic', 'cancer', 'cardiovascular', 'neurological', 'respiratory', 'autoimmune', 'other', 'unknown'));

ALTER TABLE GENE_DISEASE_ASSOCIATION

ADD CONSTRAINT GDA_GENE_FK

FOREIGN KEY (gene_id) REFERENCES GENE (gene_id)

ON DELETE CASCADE ON UPDATE CASCADE,

ADD CONSTRAINT GDA_GENEDISEASE_FK

FOREIGN KEY (disease_id) REFERENCES GENE_DISEASE (disease_id)

ON DELETE CASCADE ON UPDATE CASCADE;

ALTER TABLE GENE_PATHWAY

ADD CONSTRAINT GENEPATH_NAME_UNIQUE

UNIQUE (pathway_name);

ALTER TABLE GENE_PATHWAY_ASSOCIATION

ADD CONSTRAINT GPA_GENE_FK

FOREIGN KEY (gene_id) REFERENCES GENE (gene_id)

ON DELETE CASCADE ON UPDATE CASCADE,

ADD CONSTRAINT GPA_PATHWAY_FK

FOREIGN KEY (pathway_id) REFERENCES GENE_PATHWAY (pathway_id)

ON DELETE CASCADE ON UPDATE CASCADE;

ALTER TABLE GENE_FAMILY

ADD CONSTRAINT GENEFAMILY_NAME_UNIQUE

UNIQUE (family_name);


```

ALTER TABLE GENE_FAMILY_MEMBER
ADD CONSTRAINT GFM_GENE_FK
    FOREIGN KEY (gene_id) REFERENCES GENE (gene_id)
    ON DELETE CASCADE ON UPDATE CASCADE,
ADD CONSTRAINT GFM_FAMILY_FK
    FOREIGN KEY (family_id) REFERENCES GENE_FAMILY(family_id)
    ON DELETE CASCADE ON UPDATE CASCADE;

```

```

ALTER TABLE GO_TERM
ADD CONSTRAINT GOTERM_NAME_UNIQUE
    UNIQUE(term_name),
ADD CONSTRAINT GOTERM_ONTOLOGY_CHECK
    CHECK(ontology in ('cc', 'mf', 'bp')),
ADD CONSTRAINT GOTERM_UPDATE_CHECK
    CHECK(date_last_update >= date_created),
ADD CONSTRAINT GOTERM_ID_CHECK
    CHECK(go_id REGEXP '^GO:[0-9]{7}$');

```

```

ALTER TABLE GO_TERM_RELATION
ADD CONSTRAINT GTR_PARENT_FK
    FOREIGN KEY (parent_go_id) REFERENCES GO_TERM (go_id)
    ON DELETE RESTRICT ON UPDATE CASCADE,
ADD CONSTRAINT GTR_CHILD_FK
    FOREIGN KEY (child_go_id) REFERENCES GO_TERM (go_id)
    ON DELETE RESTRICT ON UPDATE CASCADE,
ADD CONSTRAINT GTR_TERM_CHECK
    CHECK(relation_term IN ('is a', 'part of', 'has part', 'regulates', 'occurs in'));

```

```

ALTER TABLE GENE_ANNOTATION
ADD CONSTRAINT GENEANN_GENE_FK
    FOREIGN KEY (gene_product_id) REFERENCES GENE_PRODUCT (gene_product_id)
    ON DELETE RESTRICT ON UPDATE CASCADE,
ADD CONSTRAINT GENEANN_GOTERM_FK
    FOREIGN KEY (go_id) REFERENCES GO_TERM (go_id)
    ON DELETE RESTRICT ON UPDATE CASCADE,
ADD CONSTRAINT GENEANN_CODE_CHECK
    CHECK(evidence_code IN ('EXP', 'IDA', 'IPI', 'IMP', 'IGI', 'IEP', 'HTP', 'HDA', 'HMP', 'HGI', 'HEP', 'IBA',
'IBD', 'IKR', 'IRD', 'ISS', 'ISO', 'ISA', 'ISM', 'IGC', 'RCA', 'TAS', 'NAS', 'IC', 'ND', 'IEA')),
ADD CONSTRAINT GENEANN_GENE_GO_UNIQUE
    UNIQUE(gene_product_id, go_id),
ADD CONSTRAINT GENEANN_UPDATE_CHECK
    CHECK(date_last_update >= date_created);

```

```

ALTER TABLE ANNOTATION_SOURCE
ADD CONSTRAINT ANNSOURCE_SOURCE_FK
    FOREIGN KEY(source_id) REFERENCES SOURCE(source_id)
    ON DELETE CASCADE ON UPDATE CASCADE,
ADD CONSTRAINT ANNSOURCE_ANNOTATION_FK
    FOREIGN KEY(annotation_id) REFERENCES GENE_ANNOTATION(annotation_id)
    ON DELETE CASCADE ON UPDATE CASCADE;

```

SELECT STATEMENTS

SELECT * FROM GENE;

SELECT * FROM ORGANISM;

SELECT * FROM GENE_DISEASE;

SELECT * FROM GENE_DISEASE_ASSOCIATION;

SELECT * FROM GENE_FAMILY;

SELECT * FROM GENE_FAMILY_MEMBER;

SELECT * FROM GENE_PATHWAY;

SELECT * FROM GENE_PATHWAY_ASSOCIATION;

SELECT * FROM GENE_PRODUCT;

SELECT * FROM GENE_ANNOTATION;

SELECT * FROM ANNOTATION_SOURCE;

SELECT * FROM SOURCE;

SELECT * FROM GO_TERM;

SELECT * FROM GO_TERM_RELATION;

The test data using the SELECT statements is shown below

1 SELECT * FROM GENE;							
100% 20:1							
Result Grid Filter Rows: Search Edit: Export/Import:							
gene_id	gene_symbol	short_descripti...	chromosome	start_positi...	end_positi...	strand	organism_id
1	TP53	NULL	17	7668421	7687490	-	1
2	EGFR	NULL	7	55019017	55211628	+	1
3	APOE	NULL	19	44905796	44909393	+	1
4	TNF	NULL	6	31575565	31578336	+	1
5	IL6	NULL	7	22727200	22731998	+	1
6	VEGFA	NULL	6	43770211	43786487	+	1
7	TGFB1	NULL	19	41330323	41353922	-	1
8	MTHFR	NULL	1	11785723	11805964	-	1
9	BRCA1	NULL	17	43044295	43170327	-	1
10	ACE	NULL	17	63477061	63498373	+	1
11	STAT3	NULL	17	42313324	42388442	-	1
12	Blm	NULL	3R	11736810	11741783	-	2
13	N	NULL	X	3134870	3172221	+	2
14	dpp	NULL	2L	2428372	2459823	+	2
15	PHYB	NULL	2	8139756	8144461	+	3
16	FLC	NULL	5	3173382	3179448	-	3
17	PHYA	NULL	1	3095160	3100819	-	3
18	recA	NULL	1	2822708	2823769	-	4
19	dnaA	NULL	1	3882326	3883729	-	4
20	rho	NULL	1	3966417	3967676	+	4
21	daf-16	NULL	1	10750498	10776703	+	5
22	daf-2	NULL	3	2994514	3040846	-	5
23	skn-1	NULL	4	5651039	5660384	-	5
24	BRCA2	NULL	13	32315077	32400268	+	1
25	PALB2	NULL	16	23603165	23641310	-	1
26	TRP53	NULL	11	69471174	69482699	+	6
27	TRP63	NULL	16	25502513	25710842	+	6
28	TRP73	NULL	4	154140706	154224332	-	6
29	TP63	NULL	3	189596746	189897276	+	1
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

1 SELECT * FROM ORGANISM;						
100% 24:1						
Result Grid Filter Rows: Search Edit: Export/Import:						
organism_id	scientific_name	domain	genome_size	chromosome_co...	taxonomy_id	
1	Homo sapiens	eukarya	3200000000	46	9606	
2	Drosophila melanogaster	eukarya	140000000	8	7227	
3	Arabidopsis thaliana	eukarya	135000000	10	3702	
4	Escherichia coli	bacteria	4600000	1	562	
5	Caenorhabditis elegans	eukarya	100000000	12	6239	
6	Mus musculus	eukarya	2700000000	20	10090	
NULL	NULL	NULL	NULL	NULL	NULL	

1

SELECT * FROM GENE_DISEASE;

100%

28:1

Result Grid

Filter Rows:

Search

	disease_id	disease_name	category	
	1	li-fraumeni syndrome	genetic	
	2	breast cancer	cancer	
	3	lung cancer	cancer	
	4	alzheimer's disease	neurological	
	5	cardiovascular disease	cardiovascular	
	6	rheumatoid arthritis	autoimmune	
	7	crohn's disease	autoimmune	
	8	psoriasis	autoimmune	
	NULL	NULL	NULL	

1

SELECT * FROM GENE_DISEASE_ASSOCIATION;

100%

40:1

Result Grid

Filter Rows:

Search

Edit:

	gene_id	disease_id
1	1	
1	2	
9	2	
24	2	
25	2	
1	3	
3	4	
3	5	
4	6	
4	7	
4	8	
NULL	NULL	

1SELECT * FROM GENE_FAMILY;

100%27:1

Result GridFilter Rows: Search

family_id	family_name	description
1	p53 family	NULL
2	ErbB family	NULL
3	PDGF/VEGF growth factor family	NULL
NULL	NULL	NULL

1SELECT * FROM GENE_FAMILY_MEMBER;

100%34:1

Result GridFilter Rows: Search

gene_id	family_id
1	1
26	1
27	1
28	1
29	1
2	2
6	3
NULL	NULL

1SELECT * FROM GENE_PATHWAY;

100%28:1

Result GridFilter Rows: Search

	pathway_id	pathway_name	description
	1	cell cycle pathway	NULL
	2	apoptosis pathway	NULL
	3	MAPK signaling pathway	NULL
	NULL	NULL	NULL

1SELECT * FROM GENE_PATHWAY_ASSOCIATION;

100%40:1

Result GridFilter Rows: SearchEdit:

	gene_id	pathway_id	
	1	1	
	9	1	
	1	2	
	4	2	
	7	2	
	11	2	
	1	3	
	4	3	
	5	3	
	6	3	
	7	3	
	11	3	
	NULL	NULL	

1 **SELECT * FROM GENE_PRODUCT;**

100% 28:1

Result Grid Filter Rows: Search Edit: Export/Import:

	gene_product...	gene_id	product_name	product_ty...	description
1		4	tumor necrosis factor	protein	cytokine
2		10	ACE protein	protein	angiotensin I converting enzyme
3		20	Rho protein	protein	transcription termination factor
4		9	BRCA1 protein	protein	nuclear phosphoprotein
5		24	BRCA2 protein	protein	nuclear DNA repair protein
6		25	PALB2 protein	protein	partner and localizer of BRCA2 protein
7		15	phytochrome B	protein	a plant photoreceptor that forms a photobody
8		16	FLC protein	protein	a MADS-box transcription factor that represses floral...
9		17	phytochrome A	protein	a photoreceptor responsible for sensing far-red light
	NULL	NULL	NULL	NULL	NULL

1 **SELECT * FROM GENE_ANNOTATION;**

100% 31:1

Result Grid Filter Rows: Search Edit: Export/Import:

	annotation_id	gene_product...	go_id	evidence_code	date_created	date_last_update	comment
1	1		GO:0006915	IDA	2026-04-09 16:38:52	2026-04-09 16:38:52	TNF induces apoptosis via TNFR1 signaling
2	2		GO:0007165	IDA	2026-04-09 16:38:52	2026-04-09 16:38:52	ACE catalyzes angiotensin I to II, which activate...
3	3		GO:0009987	IDA	2026-04-09 16:38:52	2026-04-09 16:38:52	Rho protein mediates the termination of transcri...
5	8		GO:0005634	EXP	2026-04-09 16:38:52	2026-04-09 16:56:58	FLC functions as a nuclear transcription factor t...
6	9		GO:0005737	IEA	2026-04-09 16:38:52	2026-04-09 16:38:52	PHYA is localized in the cytoplasm prior to bein...
	NULL	NULL	NULL	NULL	NULL	NULL	NULL

1 **SELECT * FROM ANNOTATION_SOURCE;**

100% 33:1

Result Grid Filter Rows: Search

	source_id	annotation_id
1	1	
2	2	
3	3	
5	5	
6	6	
	NULL	NULL

1

SELECT * FROM SOURCE;

100%

22:1

Result Grid



Filter Rows:

Edit:



Export/Import:



	source_id	source_name	publish_year	url
	1	Reactive Oxygen Species in TNFα-Induced Sig...	2010	https://www.sciencedirect.com/science/article/pii...
	2	Physiology, Renin Angiotensin System	2026	https://www.ncbi.nlm.nih.gov/books/NBK470410/
	3	Regulation of Rho-Dependent Transcription Ter...	2000	https://pubs.acs.org/doi/10.1021/bi992658z
	5	Light-Regulated Nuclear Import and Degradatio...	2010	https://pmc.ncbi.nlm.nih.gov/articles/PMC30370...
	6	UniProt	2024	https://www.uniprot.org/uniprotkb/Q9S7Z8/entry
	NULL	NULL	NULL	NULL

1SELECT * FROM GO_TERM;

100%23:1

Result Grid

Filter Rows:

Search

Edit:

Export/Import:

go_id	term_name	ontology	definition	date_created	date_last_update
GO:0005634	nucleus	cc	A membrane bound organelle in eukaryotic cells...	2026-04-09 16:38:52	2026-04-09 16:38:52
GO:0005737	cytoplasm	cc	The fluid-like substance that fills the interior of t...	2026-04-09 16:38:52	2026-04-09 16:42:38
GO:0006915	apoptotic process	bp	The process of programmed cell death that occ...	2026-04-09 16:38:52	2026-04-09 16:38:52
GO:0007154	cell communication	bp	Any process by which a cell gives and receive i...	2026-04-09 16:38:52	2026-04-09 16:38:52
GO:0007165	signal transduction	bp	The cellular process where a signal is transmitt...	2026-04-09 16:38:52	2026-04-09 16:38:52
GO:0007267	cell-cell signaling	bp	Any process that mediates the transfer of inform...	2026-04-09 16:38:52	2026-04-09 16:38:52
GO:0008283	cell proliferation	bp	The process by which cells grow and divide to c...	2026-04-09 16:38:52	2026-04-09 16:38:52
GO:0009987	cellular process	bp	Any process that is carried out at the cellular level.	2026-04-09 16:38:52	2026-04-09 16:38:52
GO:0010647	positive regulation of cell communication	bp	A process that increases the rate of cell commu...	2026-04-09 16:38:52	2026-04-09 16:38:52
GO:0012501	programmed cell death	bp	A highly regulated process of cellular self-destru...	2026-04-09 16:38:52	2026-04-09 16:38:52
GO:0035426	extracellular matrix-cell signaling	bp	Any process that mediates the transfer of inform...	2026-04-09 16:38:52	2026-04-09 16:38:52
NULL	NULL	NULL	NULL	NULL	NULL

1

SELECT * FROM GO_TERM_RELATION;

100%32:1

Result GridFilter Rows:

Search

	go_relation_id	parent_go_id	child_go_id	relation_ter...
	1	GO:0009987	GO:0007165	is a
	2	GO:0009987	GO:0008283	is a
	3	GO:0012501	GO:0006915	is a
	4	GO:0007154	GO:0007165	part of
	5	GO:0007154	GO:0007267	is a
	6	GO:0007154	GO:0035426	is a
	7	GO:0007154	GO:0010647	regulates
	NULL	NULL	NULL	NULL